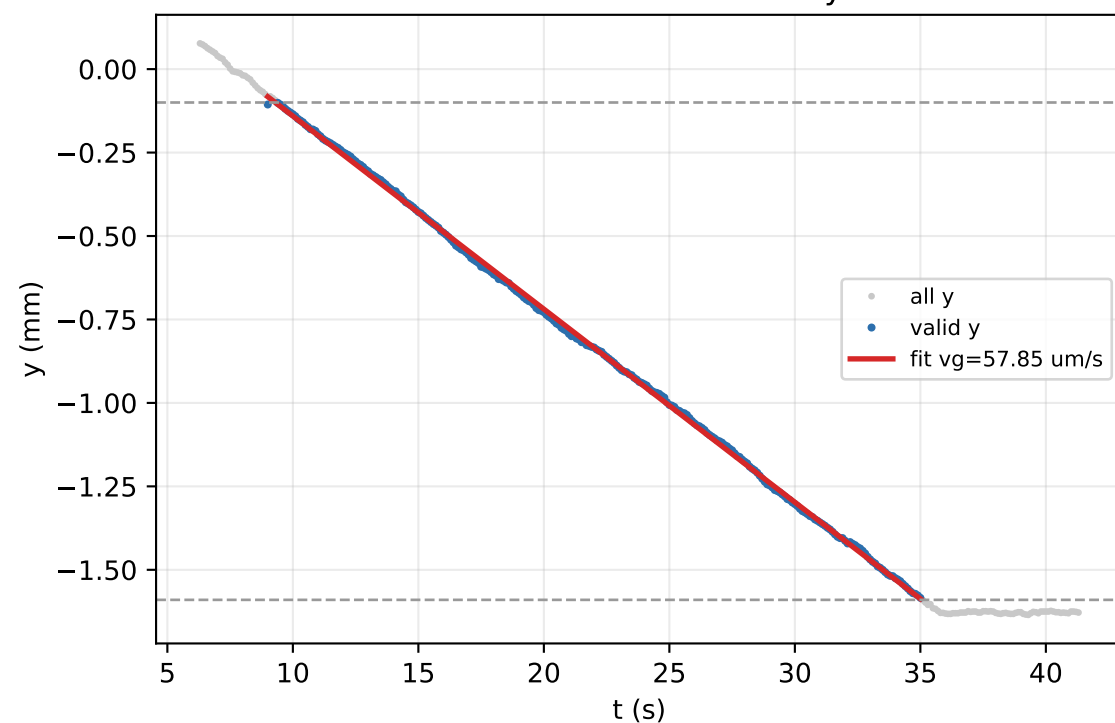
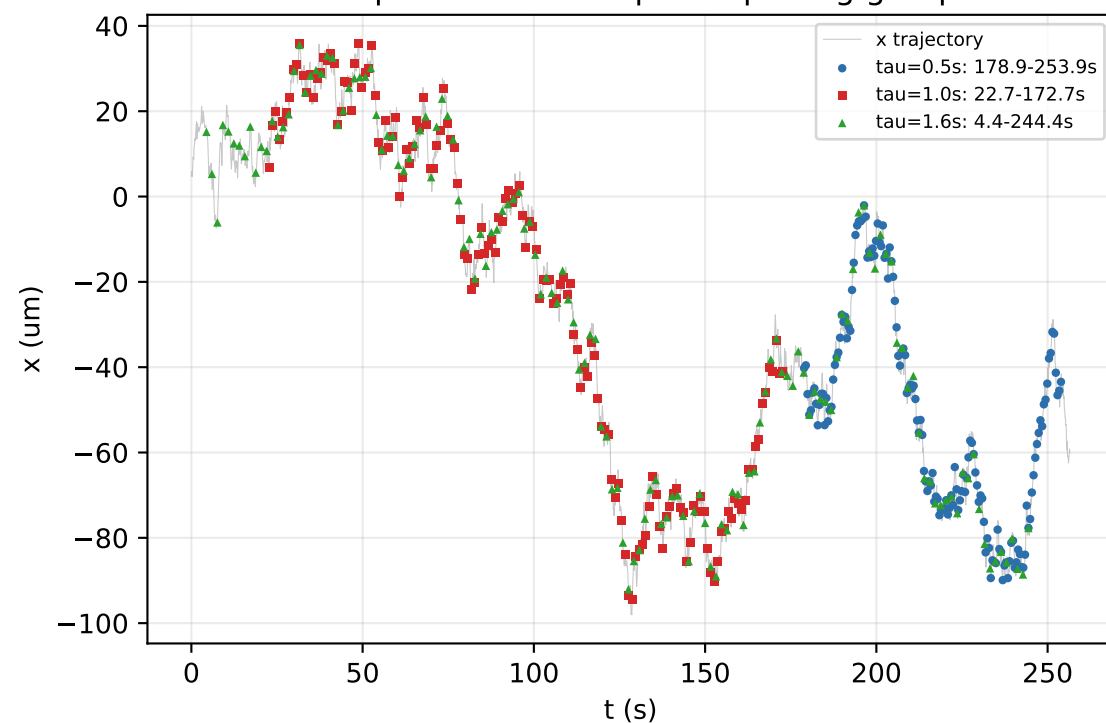


Millikan oil drop particle2: optimized Brownian intervals

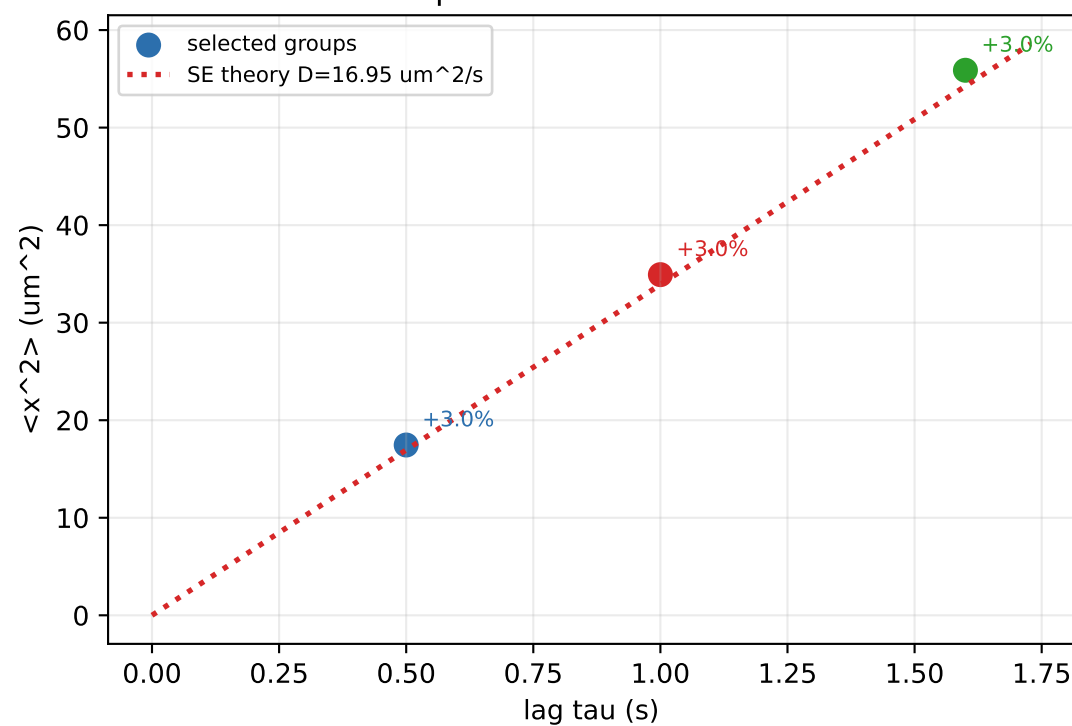
Free-fall terminal velocity



Sampled coordinate points per lag group



Separate $\langle x^2 \rangle$ results



Summary

Particle: particle2
Window strategy: target_kb
Freefall valid points: 258
 $vg = 57.8525 \text{ um/s}$, $R^2 = 0.999677$
 $r_{\text{eff}} = 0.7019 \text{ um}$, $d_{\text{eff}} = 1.4038 \text{ um}$
SE theory $D = 16.9534 \text{ um}^2/\text{s}$

tau	covered time	$\langle x \rangle$	$\langle x^2 \rangle$	k_B err
0.5s	178.9-253.9s	-0.022	17.458	+2.98%
1.0s	22.7-172.7s	-0.318	34.925	+3.00%
1.6s	4.4-244.4s	-0.619	55.878	+3.00%

NIST $k_B = 1.3806\text{e-}23 \text{ J/K}$
No average is reported.